

Power Budget: Working Document

Digital Audio Player • Rev 7

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1. Summary	1
2. Design Requirements	2
3. Source Data	2
4. Conversion Efficiency	2
5. Operating Scenarios	3
5.1 S1: Standby	3
5.2 S2: Playback, display off (dominant)	3
5.3 S3: Browsing, display on	3
5.4 S4: Absolute peak	4
6. Battery Sizing	4
7. Charging	4
8. Regulator and Thermal Sizing	5
9. Confidence Register	5
10. Validation Plan	5
11. Revision Log	5
12. Sources	6

1. Summary

Result	Value
Playback, display off (S2)	≈ 605 mW (164 mA at 3.7 V)
Weighted average (90 % S2, 10 % S3)	≈ 641 mW (173 mA)
Runtime on derated capacity	≈ 7.8 h against a 7 h target
Absolute peak (S4)	≈ 1.49 W (403 mA at 3.7 V)
Standby (S1)	≈ 18 mW (≈ 16 days to flatten the cell)

2. Design Requirements

Requirement	Value
Battery life	≥ 7 h continuous playback, derated, 90 % S2 / 10 % S3
Maximum audio output	0.6 V RMS into the EW300 (28 Ω ±15 %)
Cell	Jauch LP605060JU+PCM, 1850 mAh (6.8 Wh)
Charge current	500 mA (0.27C)
Constraints	Low noise · Small footprint

3. Source Data

Block	Active	Passive	Rail	Basis
DAC (ES9038Q2M)	53 mW	5 mW	+3V3_ADC / +3V3_DIG	26 mW analog (AVCC 6 mA, VCCA 2 mA), 26 mW digital
MCU (ESP32-S3)	165 mW	7 μW	+3V3_DIG	Datasheet
microSD	330 mW	3 mW	+3V3_DIG	Typical; not power-gated
Oscillator (one active)	7 / 18 mW	≈ 0	+3V3_ANALOG	ECS 2.2 mA / Abracon 5.5 mA max
OPA1622 (2 ch)	185 mW max	5 μA/ch	±3.3 V	Full scale into 23.8 Ω + 34 mW quiescent
OPA1612 (6 ch)	143 mW	No shutdown	±3.3 V	3.6 mA/ch × 6 × 6.6 V
Display + backlight	300 mW	≈ 0	+3V3_DIG + SYS	Backlight 78 mA × 3.2 V ≈ 250 mW at bright via BD1604MUV (1x mode ≈ 86 % at 3.7 V); dim 19 mA ≈ 72 mW
Charger leakage	n/a	60 to 80 μW	BAT	16 μA typ, 22 μA max

$$P_{supply} = \frac{2V_S I_{pk}}{\pi} = \frac{2 \times 3.3 \text{ V} \times 35.7 \text{ mA}}{\pi} = 75 \text{ mW} \Rightarrow 2 \times 75 + 34 \approx 185 \text{ mW} \quad (1)$$

Source: [1], [2], [3], [4], [8]

4. Conversion Efficiency

Branch	Topology	Efficiency	Note
Charger to SYS	BQ24250, 3 MHz buck	90 to 93 %	Datasheet curves
Digital 3.3 V	LTC3440, 600 kHz	≈ 90 %	Switcher only
Analog +3.3 V (two LT3042)	LTC3440 3.9 V, then LDO	≈ 77 %	≈ 90 % × ≈ 85 %
Analog -3.3 V	LTC3440, LT3462A, LT3093	≈ 52 to 55 %	LT3462A simulated 67.7 % at 98 mA
Backlight	BD1604MUV charge pump from SYS	≈ 76 to 86 % (1x)	V_f / V_SYS; 1.5x mode only near empty

5. Operating Scenarios

5.1 S1: Standby

Block	Draw	Basis
DAC	$\leq 5 \text{ mW}$	Standby at DVCC = 3.3 V
MCU	$< 0.1 \text{ mW}$	Deep sleep
microSD	$\approx 3 \text{ mW}$	Idle; no load switch
Analog branch	≈ 0	3.9 V LTC3440 off; inverter and LDOs follow
Digital LTC3440	$\approx 8 \text{ mW}$	Fixed frequency, always on: $I_Q = f(0.75 V_{IN} + 0.25 V_{OUT}) \approx 2.2 \text{ mA}$ at 3.7 V
Static pull-ups	$\approx 1.3 \text{ mW}$	Encoder A closed at rest 0.33 mA worst case; R221 33 μA ; plug detect 30 μA
Backlight, display	≈ 0	BD1604MUV EN low; panel in sleep
Charger leakage	$\approx 0.08 \text{ mW}$	BAT
Total S1	$\approx 18 \text{ mW}$	≈ 16 days from 6.8 Wh

5.2 S2: Playback, display off (dominant)

Block	Effective draw	Basis
MCU	165 mW	Decode and I2S
DAC	53 mW	Active
microSD	$\approx 10 \text{ mW}$	2 % duty
Oscillator	$\approx 18 \text{ mW}$	Worst case
OPA1622	$\approx 50 \text{ mW}$	Quiescent + typical signal
OPA1612 (6 ch)	143 mW	Quiescent
Display	$\approx 15 \text{ mW}$	5 % duty
Load subtotal	$\approx 454 \text{ mW}$	
Conversion losses	$\approx 151 \text{ mW}$	33 % loss ratio
Total S2	$\approx 605 \text{ mW}$	164 mA at 3.7 V

5.3 S3: Browsing, display on

Block	Effective draw
MCU	165 mW
DAC + analog chain	230 mW
Oscillator	$\approx 18 \text{ mW}$
microSD	$\approx 40 \text{ mW}$
Display + backlight	300 mW
Conversion losses	$\approx 208 \text{ mW}$
Total S3	$\approx 961 \text{ mW}$ (260 mA)

5.4 S4: Absolute peak

Block	Peak draw
MCU	165 mW
DAC	53 mW
microSD	330 mW
Oscillator	18 mW
OPA1622	185 mW
I/V and summing	≈ 155 mW
Display + backlight	300 mW
Conversion losses	≈ 288 mW
Total S4	≈ 1.49 W (403 mA at 3.7 V)

6. Battery Sizing

$$P_{avg} = 0.90 \times 605 + 0.10 \times 961 = 641 \text{ mW}, \quad I_{avg} = 641 \text{ mW} / 3.7 \text{ V} = 173 \text{ mA} \quad (2)$$

Factor	Multiplier
Usable depth of discharge	0.90
End-of-life capacity fade	0.85
Low-temperature derating	0.95
Combined	0.73: 1850 mAh × 0.73 = 1350 mAh usable

Condition	Current	Runtime
Weighted average, nameplate	173 mA	10.7 h
Weighted average, derated (working figure)	173 mA	7.8 h
With 20 % design margin	208 mA	6.5 h
Continuous browsing	260 mA	5.2 h

7. Charging

Parameter	Value	Note
Charge current	500 mA (0.27C)	R_ISET 500 Ω
Charge time	≈ 4.5 to 5 h	Including CV taper
Safety timer	9 h, set via I2C	Required configuration step
Input limit	1 A (EN1 high, EN2 low); 500 mA while +3V3_DIG is down	V_IN_DPM 4.488 V on weak sources
Charger dissipation	≈ 160 mW, ≈ 5 °C rise	θ_JA 32.9 °C/W

Source: [5]

8. Regulator and Thermal Sizing

Item	Result
Charger inductor	$403 + 500 + 222 \text{ mA} \approx 1.13 \text{ A}$ vs 2 A saturation
LT3042 U204 (+3V3_ANALOG)	$\approx 32 \text{ mA}$ quiescent (OPA1612 21.6 + OPA1622 5.2 + oscillator ≤ 5.5) + 71.3 mA signal $\approx 104 \text{ mA}$ vs 125 mA limit
LT3042 DAC LDO (+3V3_ADC)	$\approx 8 \text{ mA}$ (AVCC 6 + VCCA 2)
LT3093 (-3.3 V)	$\approx 27 \text{ mA}$ quiescent + 71.3 mA signal $\approx 98 \text{ mA}$ vs 126.6 mA limit
LT3462A input	$\approx 147 \text{ mA}$ at 98 mA output
LTC3440 (SYS $\rightarrow$ 3.9 V)	$\approx 263 \text{ mA}$ (both LT3042 incl. $\approx 2 \text{ mA}$ ground current each, plus LT3462A); $\approx 5 \%$ over the 250 mA design current
LTC3440 (SYS $\rightarrow$ 3.3 V)	$\approx 260 \text{ mA}$ peak vs 300 mA design current
BD1604MUV	$\approx 80 \text{ mA}$ LED at bright from SYS; $4 \times 20 \text{ mA}$ within its $4 \times 30 \text{ mA}$ rating

Source: [6], [7], [8], [9]

9. Confidence Register

Item	Status	Action
Display + backlight (300 mW)	Estimate; backlight now calculated	Measure the panel logic and LED current at operating brightness.
OPA1612 / OPA1622 quiescent	Datasheet typical	Confirm at $\pm 3.3 \text{ V}$.
ES9038Q2M (53 mW)	Datasheet	Confirm the power mode in use.
microSD figures	Typical	Measure, including idle current (no load switch).
Regulator quiescent	Calculated for the digital LTC3440	Sum actual I_Q for every regulator.

10. Validation Plan

Instrument battery current with a sense resistor in the cell return over a full discharge; log firmware state alongside; measure a full charge cycle and the TS trip temperatures; measure each regulator branch through its test points; measure standby current with the encoder at rest in both detent states.

11. Revision Log

Rev	Date	Change
3 to 6	to 2026-09-27	Earlier revisions (see Rev 6).
7	2026-10-03	Rail allocation from the complete schematic: DAC analog on a dedicated LT3042, oscillators counted on +3V3_ANALOG ($\approx 104 \text{ mA}$ peak), 3.9 V load $\approx 263 \text{ mA}$. Standby recomputed $\approx 18 \text{ mW}$ (no SD load switch, always-on digital LTC3440, static pull-ups). Backlight basis: BD1604MUV charge pump from SYS.

12. Sources

1. ES9038Q2M Datasheet v1.4 (ESS Technology)
2. OPA1612 Product Page and Datasheet (Texas Instruments)
3. OPA1622 Datasheet, SBOS727B (Texas Instruments)
4. ECS-2520MVL / ASTX-H11 Datasheets
5. bq2425x Product Page and Datasheet (Texas Instruments)
6. LTC3440 Product Page and Datasheet (Analog Devices)
7. LT3042 / LT3093 / LT3462A Datasheets (Analog Devices)
8. [BD1604MUV / BD1604MVV Datasheet \(ROHM\)](#)
9. Subsystem working documents (current revisions)